NIOS Class 10 (212) Science & Technology — Master Formula & Concept Cheat Sheet

NIOS Secondary (Class 10) Science & Technology (212) — Exam Prep Study Kit

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> **Keep this file open on your phone or print it out.**
> Over 65% of exam marks come from direct application of these formulas, chemical equations, and biological definitions.

PART 1: PHYSICS FORMULAS & SI UNITS

1. Motion, Force & Gravitation

- **Newton's Second Law**:

$$F = ma = \frac{m(v-u)}{t}$$

(Force F in Newtons N, mass m in kg, acceleration a in $\mathrm{m/s^2}$)

- **Momentum**:

$$p = mv \quad (\text{SI Unit: } \mathrm{kg \cdot m/s})$$

- **Universal Law of Gravitation**:

$$F = G\frac{m_1 m_2}{r^2}$$

($G = 6.67 \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$ is the Universal Gravitational Constant)

- **Acceleration due to gravity (g)**:

$$g = \frac{GM}{R^2} \approx 9.8 \text{ m/s}^2 \quad (\text{On Moon, } g_{\text{moon}} = \frac{1}{6} g_{\text{earth}})$$

- **Weight**:

$$W = mg \quad (\text{Weight is a force, measured in Newtons N})$$

- **Archimedes' Principle**: $\text{Buoyant Force} = \text{Weight of fluid displaced.}$

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2. Work, Energy & Power

- **Work Done**:

$$W = F \times s \quad (\text{SI Unit: Joule, J})$$

(1 Joule is the work done when a force of 1 N moves a body through a distance of 1 m)

- **Kinetic Energy**:

$$KE = \frac{1}{2}mv^2$$

- **Gravitational Potential Energy**:

$$PE = mgh$$

- **Power**:

$$P = \frac{\text{Work}}{\text{Time}} = \frac{W}{t} \quad (\text{SI Unit: Watt, W})$$

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3. Light (Optics)

- **Snell's Law of Refraction**:

$$\mu = \frac{\sin i}{\sin r} = \text{Constant}$$

- **Mirror Formula (Spherical Mirrors)**:

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u}$$

- Magnification: $m = -\frac{v}{u} = \frac{h_i}{h_o}$

- **Lens Formula (Spherical Lenses)**:

$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$

- Magnification: $m = +\frac{v}{u} = \frac{h_i}{h_o}$
- **Sign Convention (Cartesian)**:
- Object distance u is ALWAYS negative $(-)$.
- Concave mirror / Concave lens: focal length f is ALWAYS negative $(-)$.
- Convex mirror / Convex lens: focal length f is ALWAYS positive $(+)$.
- **Power of a Lens**:

$$P = \frac{1}{f \text{ (in metres)}} \quad (\text{SI Unit: Dioptre, D})$$

- Convex lens $\implies$ Positive power (e.g. $+2$ D).
- Concave lens $\implies$ Negative power (e.g. -1.5 D).
- **Human Eye Defects & Correction**:
- **Myopia (Short-sightedness)**: Can see near objects clearly, cannot see distant objects. Image forms in front of retina. Corrected using **Concave Lens** of suitable power.
- **Hypermetropia (Long-sightedness)**: Can see distant objects clearly, cannot see near objects. Image forms behind retina. Corrected using **Convex Lens** of suitable power.

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4. Electricity & Magnetism

- **Electric Current**:

$$I = \frac{Q}{t} \quad (\text{Current in Amperes A}, \ Q \text{ in Coulombs C})$$

- **Potential Difference**:

$$V = \frac{W}{Q} \quad (\text{Voltage in Volts V}, \ W \text{ in Joules J})$$

- **Ohm's Law**:

$$V = IR \implies R = \frac{V}{I} \quad (\text{Resistance in Ohms}, \Omega)$$

- **Resistance & Resistivity**:

$$R = \rho \frac{l}{A}$$

(l = length in m, A = cross-sectional area in m^2, ρ = resistivity in $\Omega \cdot \mathrm{m}$)

- **Resistors in Series**:

$$R_s = R_1 + R_2 + R_3 \quad (\text{Current } I \text{ remains same, voltage divides})$$

- **Resistors in Parallel**:

$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} \quad (\text{Voltage } V \text{ remains same, current divides})$$

- **Joule's Law of Heating**:

$$H = I^2 R t = V I t = \frac{V^2}{R} t \quad (\text{in Joules})$$

- **Electric Power**:

$$P = V I = I^2 R = \frac{V^2}{R} \quad (\text{in Watts})$$

- **Commercial Unit of Electrical Energy**:

$$1\ \mathrm{kWh} = 1\ \mathrm{Unit} = 1000\ \mathrm{W} \times 3600\ \mathrm{s} = 3.6 \times 10^6\ \mathrm{Joules}$$

- **Domestic Circuit Wires**:
- Live Wire: Red or Brown (carries current at $220\ \mathrm{V}$)
- Neutral Wire: Black or Light Blue (completes circuit at $0\ \mathrm{V}$)
- Earth Wire: Green or Yellow (safety measure connected to metal casing)
- **Fleming's Left-Hand Rule (Electric Motor)**:
- **Thumb**: Direction of Force / Motion
- **Forefinger**: Direction of Magnetic Field
- **Middle Finger**: Direction of Current

PART 2: CHEMISTRY FORMULAS, EQUATIONS & SALTS

1. The 4 Essential Chemical Salts (Guaranteed 4 to 6 Marks!)

Chemical Name	Common Name	Chemical Formula	Preparation Equation	Key Practical Uses
Calcium Oxychloride	Bleaching Powder	$CaOCl_2$	$Ca(OH)_2 + Cl_2 \rightarrow CaOCl_2 + H_2O$	1. Disinfectant in drinking water 2. Bleaching cotton in textiles
Sodium Hydrogen Carbonate	Baking Soda	$NaHCO_3$	$NaCl + H_2O + CO_2 + NH_3 \rightarrow NH_4Cl + NaHCO_3$	1. Ingredient in baking powder 2. Antacid for acidity relief
Sodium Carbonate Decahydrate	Washing Soda	$Na_2CO_3 \cdot 10H_2O$	$Na_2CO_3 + 10H_2O \rightarrow Na_2CO_3 \cdot 10H_2O$	1. Softening permanent hard water 2. Glass, soap, and paper industries
Calcium Sulphate Hemihydrate	Plaster of Paris (POP)	$CaSO_4 \cdot \frac{1}{2}H_2O$	$CaSO_4 \cdot 2H_2O \xrightarrow{373\ K} CaSO_4 \cdot \frac{1}{2}H_2O + 1\frac{1}{2}H_2O$	1. Plastering fractured bones 2. Making statues and decorative casts

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2. Acids, Bases & pH Indicators

- **pH Scale**:
- $pH < 7 \implies$ Acidic (e.g. Gastric juice $pH \approx 1.2$, Lemon juice ≈ 2.2)
- $pH = 7 \implies$ Neutral (Pure water, $NaCl$ solution)
- $pH > 7 \implies$ Basic / Alkaline (Blood ≈ 7.4, Milk of Magnesia ≈ 10, $NaOH \approx 14$)
- **Indicator Color Changes**:
- **Blue Litmus**: Turns RED in acid; remains blue in base.
- **Red Litmus**: Turns BLUE in base; remains red in acid.
- **Phenolphthalein**: Colourless in acid; turns PINK in base.

- **Methyl Orange**: RED in acid; YELLOW in base.

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3. Metals & Metallurgy

- **Reactivity Series**:

$$\mathrm{K} > \mathrm{Na} > \mathrm{Ca} > \mathrm{Mg} > \mathrm{Al} > \mathrm{Zn} > \mathrm{Fe} > \mathrm{Pb} > [\mathrm{H}] > \mathrm{Cu} > \mathrm{Hg} > \mathrm{Ag} > \mathrm{Au}$$

  (Potassium and Sodium are stored under kerosene oil because they react vigorously with atmospheric oxygen and moisture)
- **Roasting vs Calcination**:
- **Roasting**: Heating sulphide ore in **excess air** below its melting point:

$$2\mathrm{ZnS} + 3\mathrm{O_2} \xrightarrow{\Delta} 2\mathrm{ZnO} + 2\mathrm{SO_2}$$

- **Calcination**: Heating carbonate ore in **absence or limited air**:

$$\mathrm{ZnCO_3} \xrightarrow{\Delta} \mathrm{ZnO} + \mathrm{CO_2}$$

- **Thermite Reaction**:

$$\mathrm{Fe_2O_3} + 2\mathrm{Al} \rightarrow 2\mathrm{Fe\ (molten)} + \mathrm{Al_2O_3} + \mathrm{Heat} \quad \text{(Used to weld railway tracks)}$$

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4. Carbon and Its Compounds

- **Why Carbon forms millions of compounds**:
- **Catenation**: Unique property of carbon atoms to form long chains with other carbon atoms.
- **Tetravalency**: Valency of 4; forms strong covalent bonds with 4 other atoms.
- **Homologous Series**:
- **Alkanes** (single bond): C_nH_{2n+2} (Methane $\mathrm{CH_4}$, Ethane $\mathrm{C_2H_6}$, Propane $\mathrm{C_3H_8}$)
- **Alkenes** (double bond): C_nH_{2n} (Ethene $\mathrm{C_2H_4}$)
- **Alkynes** (triple bond): C_nH_{2n-2} (Ethyne / Acetylene $\mathrm{C_2H_2}$)
- **Functional Groups**:
- Alcohol: $-\mathrm{OH}$ (Ethanol $\mathrm{C_2H_5OH}$)
- Aldehyde: $-\mathrm{CHO}$ (Ethanal $\mathrm{CH_3CHO}$)
- Ketone: $>\mathrm{C{=}O}$ (Propanone / Acetone $\mathrm{CH_3COCH_3}$)
- Carboxylic Acid: $-\mathrm{COOH}$ (Ethanoic / Acetic Acid $\mathrm{CH_3COOH}$)

- **Saponification (Soap Making)**:

$$\text{Ester (Fat/Oil)} + \text{NaOH} \rightarrow \text{Soap (Sodium salt of fatty acid)} + \text{Glycerol}$$

- **Soap Cleansing Action**:
- Soap molecule has two ends:
  - **Hydrophilic ionic head**: Attracted to water (points outward).
  - **Hydrophobic hydrocarbon tail**: Attracted to oil/dirt (points inward).
- Together they form spherical clusters called **Micelles**, trapping oil droplets to be rinsed away with water.

PART 3: BIOLOGY CORE CONCEPTS & GENETICS

1. Life Processes & Physiology

- **Photosynthesis Equation**:

$$6CO_2 + 12H_2O \xrightarrow[\text{Chlorophyll}]{\text{Sunlight}} C_6H_{12}O_6 + 6O_2 + 6H_2O$$

- **Breakdown of Glucose by Various Pathways**:

$$\text{Glucose (6C)} \xrightarrow{\text{in Cytoplasm}} \text{Pyruvate (3C)} + \text{Energy}$$

- (a) **In absence of O_2 (Yeast / Fermentation)**: $\rightarrow \text{Ethanol} + CO_2 + \text{Energy (2 ATP)}$
- (b) **In lack of O_2 (Muscle cells)**: $\rightarrow \text{Lactic acid} + \text{Energy (Causes muscle cramps)}$
- (c) **In presence of O_2 (Mitochondria / Aerobic)**: $\rightarrow CO_2 + H_2O + \text{Energy (38 ATP)}$
- **Double Circulation in Human Heart**:
- Blood passes through the heart **twice** in one complete cardiac cycle:
  1. **Pulmonary Circulation**: Deoxygenated blood from right ventricle $\rightarrow$ Lungs $\rightarrow$ Oxygenated blood to left atrium.
  2. **Systemic Circulation**: Oxygenated blood from left ventricle $\rightarrow$ Body organs $\rightarrow$ Deoxygenated blood to right atrium.
- **Nephron (Excretory Unit of Kidney)**:
- Composed of: Bowman's capsule, Glomerulus, Renal tubule, Collecting duct.
- Three steps of urine formation:
  1. **Ultrafiltration** (Glomerular filtration)
  2. **Selective reabsorption** (Glucose, amino acids, salts, water)
  3. **Tubular secretion**.

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2. Genetics & Heredity

- **Mendel's Monohybrid Cross (Tall $\times$ Dwarf Pea Plant)**:
- Parents: $TT \times tt$
- F_1 Generation: All Tt (All Tall)
- F_2 Generation:
  - **Phenotypic Ratio**: $\mathbf{3 : 1}$ (3 Tall : 1 Dwarf)
  - **Genotypic Ratio**: $\mathbf{1 : 2 : 1}$ ($1\,TT : 2\,Tt : 1\,tt$)
- **Sex Determination in Humans**:
- Female gametes (Eggs) carry only X chromosome ($22 + X$).
- Male gametes (Sperms) carry either X or Y chromosome ($22 + X$ or $22 + Y$).
- If sperm carrying X fertilizes egg $\rightarrow XX \implies$ **Girl Child**.
- If sperm carrying Y fertilizes egg $\rightarrow XY \implies$ **Boy Child**.
- **Conclusion**: The father's sperm determines the sex of the baby with a 50% probability for each gender.

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3. Ecology & Environment

- **10% Law of Energy Transfer (Lindeman's Rule)**:
- Only 10% of energy is transferred from one trophic level to the next.
- 90% of energy is lost as heat to the environment during metabolic processes.
- *Example*: Sun ($10,000$ J) $\rightarrow$ Plants (100 J) $\rightarrow$ Deer (10 J) $\rightarrow$ Tiger (1 J).
- **Ozone Layer Depletion**:
- Ozone (O_3) in stratosphere shields Earth from harmful UV radiation.
- Depleted by **Chlorofluorocarbons (CFCs)** from refrigerators and fire extinguishers.

$$CF_2Cl_2 \xrightarrow{UV} CF_2Cl + Cl^{\bullet}$$

$$Cl^{\bullet} + O_3 \rightarrow ClO^{\bullet} + O_2$$

- **Biomagnification**:
- Increase in concentration of non-biodegradable toxic substances (like DDT, mercury) at progressively higher trophic levels of a food chain. Highest concentration accumulates in top carnivores / humans.